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Polymer-inspired mechanical metamaterials

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Metamaterials benefit from unique architected patterns to achieve lightweight with exceptional mechanical properties inaccessible to conventional materials. Typical mechanical metamaterials are inspired by crystal-like lattice structures, whose closely packed frameworks often exhibit a rigid mechanical nature. Here, we present polymer-inspired metamaterials (PIMs) by programming deformation and strengthening mechanisms that mimic the mechanical roles of key constituent elements in polymer networks. By combining metamaterial programmability with polymer-inspired structures, we design crosslinking, proto-crystalline order, and entanglement in PIMs to enable macroscale strengthening mechanisms inspired by crosslink, molecular-density, and pre-stretch strengthening in polymers, expanding the metamaterial structure-property design space. This macroscale polymer-inspired programmability is demonstrated through PIM-based soft robotic components, including programmable humanoid tissues, robotic fingers, toes and feet, and soft biomimetic exoskeletons.

Mechanical metamaterials[1–3], created by intricate internal microstructures, exhibit mechanical properties beyond natural materials. Strategies for internal microstructures are mainly inspired by crystal-inspired structures and bonding, like body-centered-cubic (BCC)[4], face-centered-cubic (FCC)[5], or octet-truss[6], to obtain superior mechanical characteristics, such as exceptional stiffness[7–9], strength[10,11], damage tolerance[12–14], and energy absorption[15–17]. However, the internal architecture, relying on either crystal-like periodic arrangements or employing straight connections (e.g., struts or plates) similar to atomic bonding, exhibits a rigidity that limits plastic yielding, stretchability, and fracture strains and is heavily dependent on the inherent properties of parent materials to expand their attainable property space. To broaden the material property space of mechanical metamaterials, existing studies have extensively explored diverse topological designs[18–22] and introduced different strengthening mechanisms[13,23–25] for these crystal-inspired metamaterials (CIMs). Some research has also focused on biomimetic structural modifications[16,26–28] or working on their parent materials[29–32] to overcome this limit. Nevertheless, the fundamental deformation mechanics of structures with crystal-inspired bonding remain unchanged, which limits the expansion of their material properties and, consequently, their application range. In contrast, polymers, with their

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twisted, entangled, and soft molecular chains reinforced by crosslinks, exhibit a high level of plasticity and molecular mechanics different from that in crystals[33,34]. This raises a fundamental question: can mechanical metamaterials be engineered to mimic polymer-enabled behaviors, thereby changing the deformation mechanics of existing metamaterial designs?

Replicating polymer microstructures[35–37] in metamaterials, characterized by highly twisted and crosslinked molecular chains, remains challenging. This complexity is beyond regular atomic arrangements in crystal materials[38–40]. Studies[41,42] have transformed CIM-derived structures into twisted architectures to achieve enhanced flexibility and functional performance[43,44], such as fracture resistance[45], large extendability[46], extreme specific strength[47], and highly recoverable deformation[48]. These advantages inspire the development of more stretchable architected beams that are mechanically similar to flexible polymer-chain elements. Building on these features, it is of interest to move toward truly polymer-network-based structures, in which chain-based architectures are interconnected by crosslinks. In addition, it is also meaningful to design polymer-inspired chains with non-periodic and more flexible architectures to enable polymer-inspired mechanical behavior at the architectural scale. For polymer-inspired networks, a successful translation of polymer strengthening mechanisms, including molecular density (MD) strengthening, crosslink (CL) strengthening, and pre-stretch (PS) strengthening, into metamaterial design will enable the development of a type of metamaterial having polymer-like mechanical behavior for functional opportunities.

Here, we developed the polymeric structure generation (PSG) algorithm to create the polymer-inspired metamaterials (PIMs) with programmable polymer-inspired chains and crosslinks that mimic the mechanical strengthening found in polymer molecular structures (Supplementary Fig. 1 and Supplementary Movie 1). This design is inspired from the typical mechanical effects of polymer networks, in which soft chains accommodate deformation through reorientation, contour change, and progressive load transfer, while crosslinks restrict chain motion and stabilize the network. PIM displays polymer-inspired elastic, plastic, and fracture behaviors at the macroscale, filling the gap in materials Ashby charts that are currently not achievable by CIMs. Most interestingly, programmability can be achieved for PIMs thanks to architectural features that enable CL, PS, and MD strengthening. To demonstrate potential for applications, we created programmable humanoid tissues with PIMs, showcasing engineered robotic toes and fingers that mimic human mechanics. In addition, we combined inspirations from crystals and polymers to create composite CIM+PIM with significant anisotropic mechanics, enabling soft biomimetic exoskeletons and multimodal grabs with highly stretchable and energy-absorbing features.

Results and discussion

Polymer-inspired meta-structures and programmable mechanics in metamaterials

In contrast to crystal-inspired meta-structures (Fig. 1a), metamaterials inspired by polymers need to have architectural features that mimic complex molecular structures of twisted, entangled soft chains stabilized by crosslinks (Fig. 1b). To mimic such complex structures, we developed the PSG algorithm (Supplementary Note 1) to generate curved, twisted chain features that are cross-linked by straight ligaments. While the curved chains are to emulate the compliant and deformable mechanics of polymer molecular chains, the straight crosslinks connect chains to constrain the chain motion. These PIM architectural features can be described by parameters including the number of polymer-inspired chains packed per unit volume (molecular-like density, ρ), proportion of length of chains aligned with load direction (PS level, p), and the number of crosslinks per unit volume (crosslink density, n_c), Fig. 1c. Programming these parameters enables translation of chain entanglement, directional alignment, and crosslink-constrained deformation to create PIMs that have strengthening mechanisms inspired by those in polymers: (1) PS strengthening, which uses pre-aligned and partially ordered polymer-inspired chains along the loading direction to increase specific strength and stiffness; (2) MD strengthening, which exploits polymer-inspired chain entanglement within a confined volume to either restrict chain mobility and improve failure resistance or permit extremely large elastic deformation for greater flexibility; and (3) CL strengthening, which increases specific stiffness by restricting the mechanical deformation among neighboring polymer-inspired chains. Therefore, we can study to obtain in-depth understanding, then subsequently program these molecular strategies to address a question: can polymer strengthening mechanisms be translated to benefit mechanical metamaterials, and can metamaterials, in turn, provide a programmable platform to interrogate these mechanisms at a higher level of abstraction?

While the deformation mechanics of polymers at the molecular scale are well established, an equivalent framework for polymer-like deformation in architected metamaterials at the macroscale is still largely missing. This gap prevents us from combining the structural programmability of metamaterials with polymer mechanics to create artificial materials with programmable polymer-inspired functionalities. To program PIMs, the deformation and strengthening behavior of a polymer chain at the macroscale must be understood. To do so, we modeled an architectural chain inspired by polymer molecular chains, as shown in Fig. 2a, then compared the macroscale mechanical properties with the intrinsic polymer strengthening, subsequently use macroscale energy partition modeling to interpret the architectural deformation mechanisms and program the properties (see Supplementary Note 6 for details). Digital image correlation (DIC) and mechanical testing (Fig. 2b and Supplementary Movie 2) reveal macroscale responses of polymer-inspired strengthening features in PIMs. In PIMs, entanglements, PS segments, and crosslinked polymer-inspired chains locally constrain chain motion, resulted in elevated local strain captured by DIC maps in Fig. 2b.

Increasing molecular density promotes the chain entanglement (ν_{ent}), which restricts local mobility and concentrates strain within entangled zones by redirecting the energy partition from reorientation-dominated configurations toward chain-bending modes with lower reorientation energy partition ($\Phi_{reorient}$) and higher bending energy partition (Φ_{bend}). PS chains in PIMs align earlier under load, increasing the load carried per unit global strain but at the same time reducing the available non-affine accommodation and thus promoting a stiffer, less extensible response once debonding mode becomes dominant. Crosslinked regions (ν_{cl},f_{cl}) constrain the extension of polymer-inspired chain, suppressing reorientation and bending modes with reduced $\Phi_{reorient}$ and Φ_{bend} and generating localized high-strain, debonding-dominated fields where the debonding energy partition (Φ_{debond}) becomes the primary component of mechanical work (refer to Supplementary Note 6 for the model relations between the energy partitions and these strengthening events). Numerical analysis (Fig. 2c) is used to quantify how entanglements per chain (ν_{ent}), crosslinks per chain (ν_{cl}), proto-crystalline ordering (Ω), and crosslink functionality (f_{cl}) reshuffle the energy partitions (see Supplementary Note 6 for detailed and quantitative derivations).

To program the polymer-inspired strengthening in PIMs, the relations between the macroscopic energy-partitions and mechanical properties of MD, CL, and PS strengthening features designed by PSG algorithms were studied. We obtain architectures whose measured modulus E, strength σ_c, fracture energies, and fracture strains follow the programmed responses with quantitative agreement between prediction and experiment (Fig. 2d, see Supplementary Note 6 and Supplementary Figs. 2, 3 for held-out validation of model). We also experimentally compare the measured responses of polymer-inspired strengthening with intrinsic polymer strengthening, where the modulus E and strength σ_c are proportional to the

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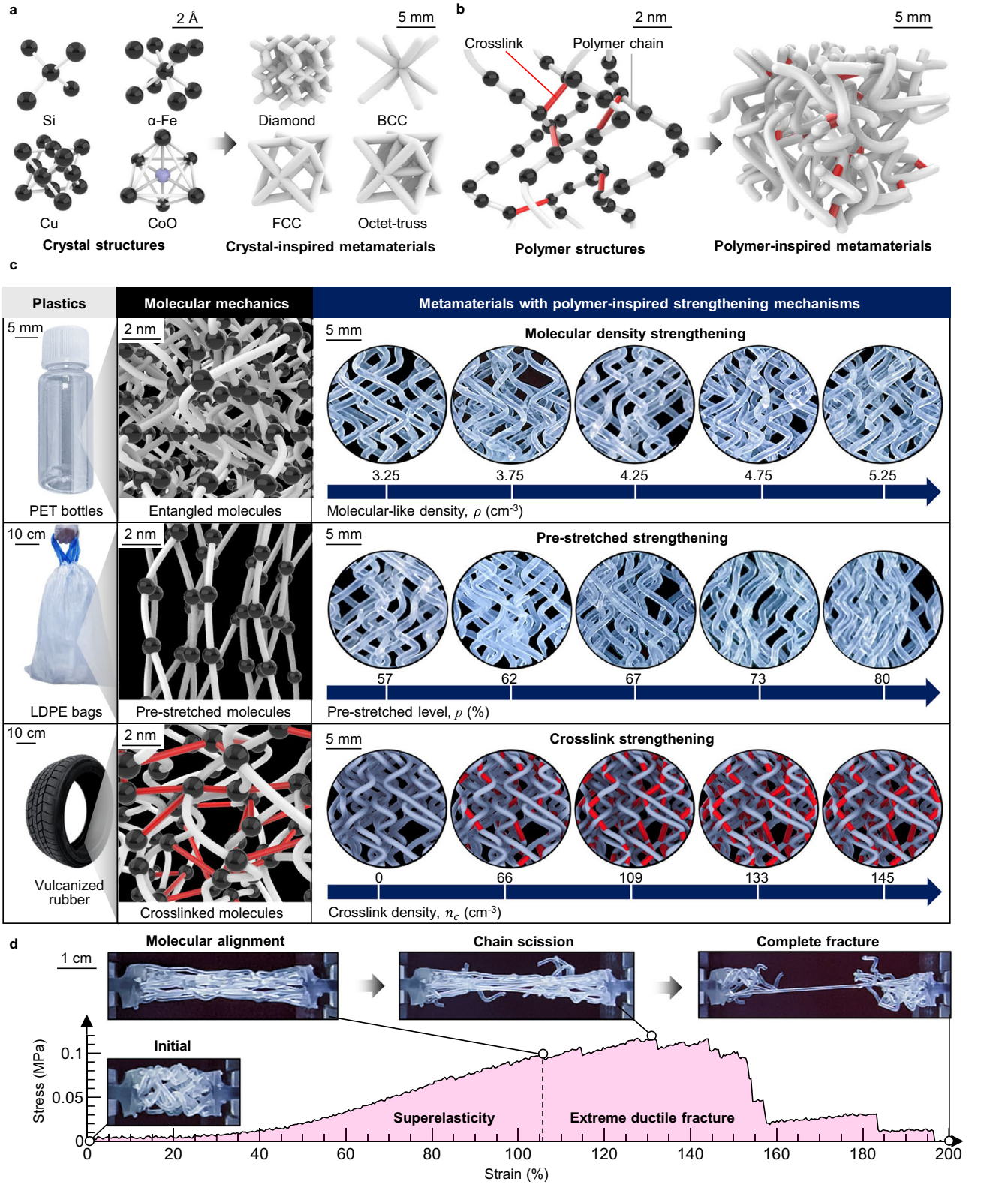


Fig. 1 | The existing CIMs and proposed PIMs achieving different polymeric strengthening mechanisms. a Typical crystal structures and existing designs of CIMs. **b** The polymer structures and PIM. **c** The PIMs mimicking MD strengthening, PS strengthening, and CL strengthening inspired from the microstructural strengthening mechanisms of polymers. The samples for molecular-density strengthening and pre-stretched strengthening are photographs of the fabricated specimens, and their colors result from the imaging conditions and do not correspond quantitatively to the arrows. The crosslink-strengthening structures are shown as renderings of our original designs created in Rhino 6, with the red color highlighting the crosslinks. The polyethylene terephthalate (PET) bottle, low-density polyethylene (LDPE) bag, and tire included in schematics are photographs of real objects captured by author Zhenyang Gao. **d** Schematic PIM stress–strain response and representative deformation sequence. Source data are provided as a Source Data file.

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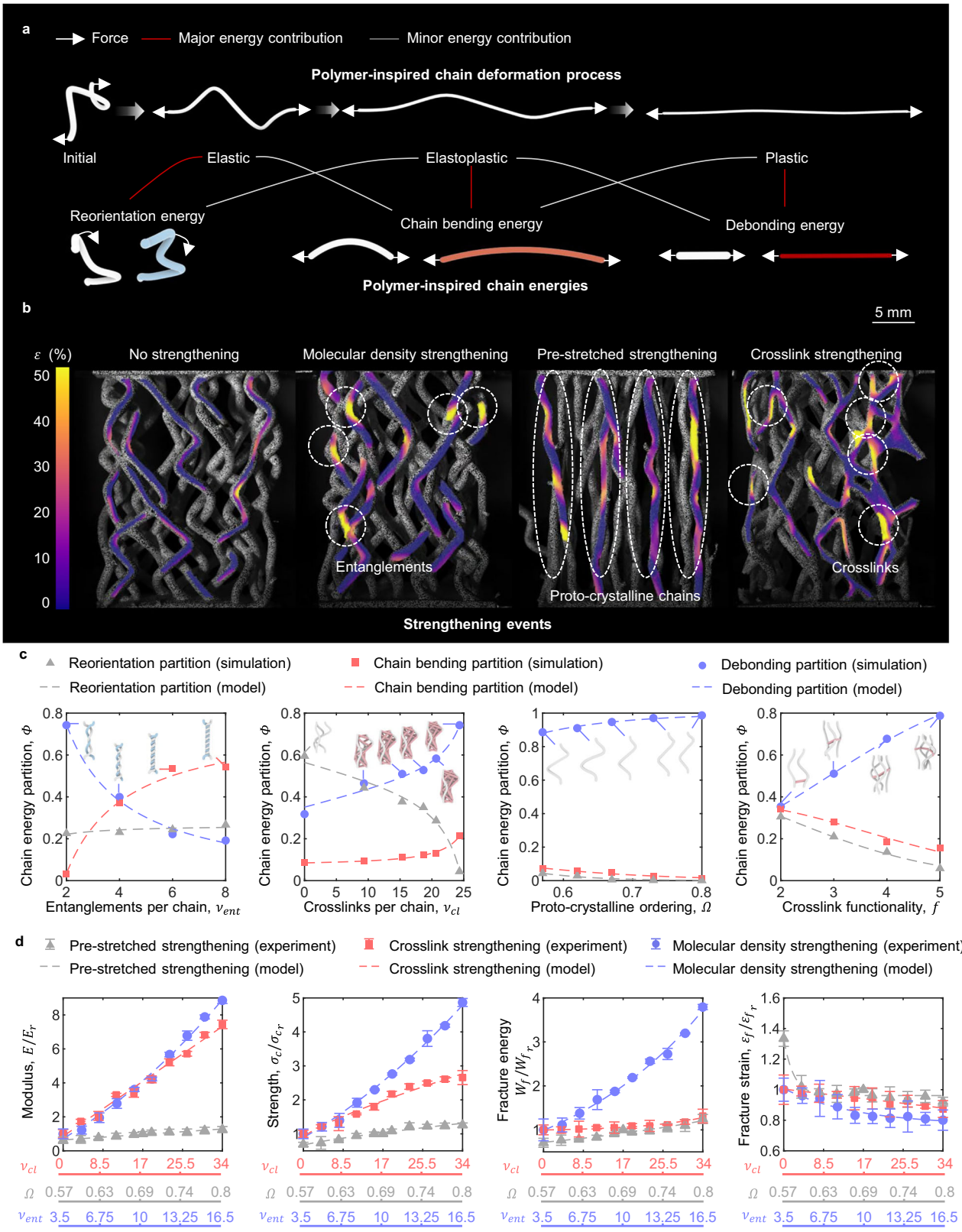


Fig. 2 | Deformation mechanics, energy partitions, and theoretical modeling of polymer-inspired metamaterials (PIMs). a Deformation of a PIM chain and its energy partitions. **b** Strain fields from digital image correlation at 15% strain for PIMs with $\rho = 4.25\,\mathrm{cm}^{-3}$, MD strengthening ($\rho = 5.25\,\mathrm{cm}^{-3}$), pre-stretched strengthening ($p = 80\%$), and CL strengthening ($n_c = 145\,\mathrm{cm}^{-3}$). ρ, p, and n_c represent the molecular density, PS level, and crosslink density, respectively. **c** Energy partitions from simulated strengthening events compared with energy partition predictions. **d** Experimental mechanical properties of PIMs and predictions from the macroscale polymer strengthening designs. The error bars are obtained as standard deviations along three repetitive experimental measurements. E_r, σ_{cr}, W_{f_r}, and ε_{f_r} are normalization modulus, strength, fracture energy, and fracture strain for experimental measure at $\nu_{cl} = 0$, $\Omega = 0.685$, and $\nu_{ent} = 3.5$. Source data are provided as a Source Data file.

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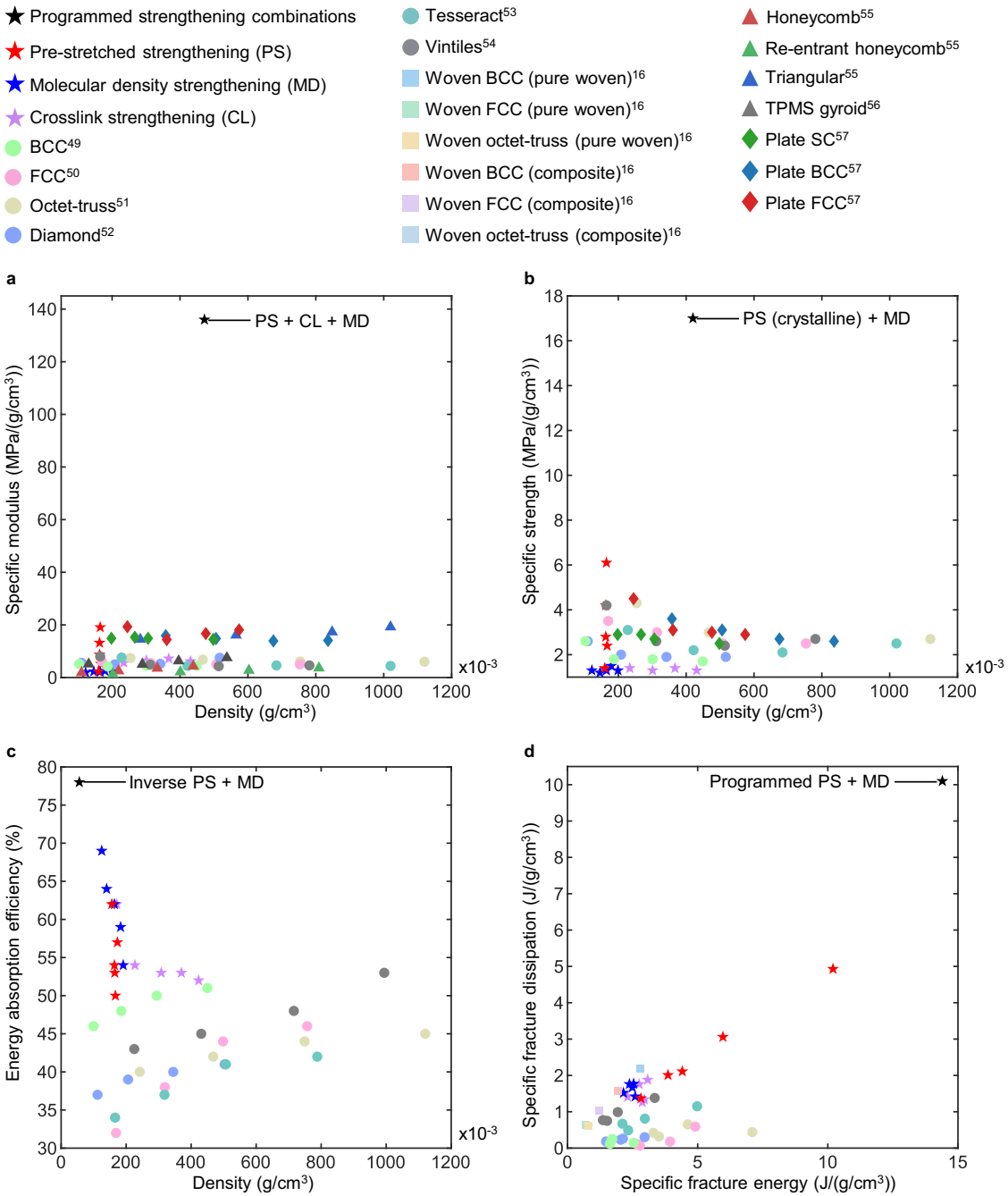


Fig. 3 | Mechanical property comparison of PIMs with representative lattice designs. The comparison architectures were based on established topological designs, including BCC[49], FCC[50], octet-truss[51], diamond[52], tesseract[53], and vintiles[54] CIMs; twisted lattice designs[16], including pure-woven and composite-woven BCC, FCC, and octet-truss structures; directional lattices[55], including honeycomb, re-entrant honeycomb, and triangular lattices; a TPMS gyroid[56]; and plate-based SC, BCC, and FCC lattices[57]. All comparison samples were designed, fabricated, and experimentally tested in this study under the same fabrication and testing conditions as PIM samples. Here, MD, PS, and CL denote MD strengthening, PS strengthening, and CL strengthening, respectively. BCC, FCC, SC, and TPMS represent body-centered cubic, face-centered cubic, simple cubic, and triply periodic minimal surface. The parent material is Formlabs® durable resin. **a** The specific modulus versus density chart. **b** The specific strength versus density. **c** The compressive energy absorption efficiency versus density. **d** The specific fracture dissipation versus specific fracture energy. Source data are provided as a Source Data file.

effective network chain density ν_e and $\sqrt{\nu_e}$, respectively, and engineering relations of polymer pre-stretch give power-law increases of λ^{n_E} and λ^{n_σ} with draw ratio λ (see Supplementary Note 7 for detailed derivations and references). These results establish a programmable mechanics framework that translates polymer strengthening mechanisms into architected metamaterials and links design parameters directly to macroscopic PIM energy partitions and mechanical responses.

Programming polymer strengthening to surpass nature

Guided by the polymer-inspired strengthening mechanisms, our results demonstrate that PIMs can be programmed to go beyond the current structural property limits (Fig. 3). Here, we demonstrate the capability of PIM through programming to improve targeted mechanical properties, including specific modulus (Fig. 3a), specific strength (Fig. 3b), energy absorption efficiency (Fig. 3c), and fracture behaviors (Fig. 3d), in comparison with CIMs (with and without twisted

structures), plate-based lattices, shell-based TPMS gyroids, and directional 2D lattices (see Supplementary Fig. 4 and Supplementary Note 2 for detailed design schematics and information). The PIM designs include CL, PS, MD strengthening, inverse PS and inverse MD (inverse PS+MD), PS with high MD (PS+MD), spatially programmed PS with high MD (programmed PS+MD), and PS, high-density crosslinked chains (PS +CL+MD).

PIMs programmed with PS macro-chains and maximized artificially engineered molecular densities were designed to favor debonding-dominated crosslinked behavior, achieving a 7× specific modulus (Fig. 3a). Meanwhile, PIMs combining polymer-inspired crystalline PS strengthening (PS (crystalline)) with MD strengthening exhibit a 4× specific strength. (Figure 3b, also see Supplementary Note 8 for crystalline PIM design). Conversely, a PIM with inverse MD and PS design is programmed to enhance chain-bending and reorientation contributions, increasing plastic plateau and resulting in a 1.5× compressive energy-absorption efficiency despite a reduction in weight (Fig. 3c). By prescribing PS distributions together with high MD to target gradient (Φ_{debond}, Φ_{bend}, $\Phi_{reorient}$) triplets, PIMs are programmed to exhibit a more ductile and progressive fracture process, absorbing 4× higher fracture energy and dissipating 5× more energy during fracture propagation relative to twisted CIMs (Fig. 3d). Our polymer-inspired strengthening approach also allows programming the mechanical behavior precisely along given orientations, including crack control (Supplementary Note 5), leading to a type of metamaterials that incorporates the mechanisms in polymers to significantly enhance and program the mechanical response, including efficient energy absorption, ultra-high modulus and strength at low weight, and improved fracture resistance. To compare the fundamental building elements of PIMs with those of strut-based metamaterials at the individual strut scale, we experimentally measured the maximum fracture strain of PIM chain elements, classical straight struts, and different helical struts under identical beam diameter and design-space conditions. Experiments show that a 1 cm PIM chain built with durable v2 resin can be drawn up to 55.8 cm, resulting in a 5480% maximum fracture strain (Supplementary Fig. 5 and Supplementary Movie 4) with a specific energy absorption of 4.8 J/g, where the fracture strain can be further extended to 6730% if fabricated by flexible 80 A resin. For a broad range of parent materials with intrinsic fracture strains $\varepsilon_f = 12\%$ (clear v4), $\varepsilon_f = 42\%$ (durable v2), and $\varepsilon_f = 63\%$ (flexible 80 A), PIM chain increased the fracture strains by 48, 25, and 18 times, respectively, compared to the helical strut elements. These results provide an elementary-scale basis for the improved programmable extendibility functions of PIMs, such as controlled fractures (Supplementary Fig. 6).

Mechanistically, in crystal-inspired lattices, straight members experience stretching/bending along well-constrained load paths, where axial stretching along struts leads to early yielding and abrupt failure. In PIMs, the architected chains spend a large portion of the deformation in reorientation- and chain-bending-dominated regimes, where deformation is accommodated by twisting, uncoiling, and contour reshaping rather than immediate backbone stretch, where the chains can be further strengthened by toughening designs (Supplementary Fig. 7). Designs, such as PS+MD and PS (crystalline)+MD deliberately suppress these responses and maximize the debonding partition Φ_{debond} over high-density chains parallelly programmed, which amplifies stiffness and strength. In contrast, the inverse PS+MD architecture promotes $\Phi_{reorient}$ and Φ_{bend} over a wide strain window, creating an extended plastic-like plateau and higher energy absorption. Artificial PS+MD distributions imprint spatially varying (Φ_{debond}, Φ_{bend}, $\Phi_{reorient}$) triplets to program a smoothly evolving resistance rather than abrupt transition between deformation regimes, giving stable, high-energy fracture dissipation. Finally, the single-chain zero-entanglement PIM pushes the entire structure through reorientation-dominated deformation before significant debonding occurs, significantly increasing the fracture strain by orders of magnitude.

Programming polymer-inspired metamaterials for humanoid tissues

Biological muscles and tendons play a pivotal role in linking bones and transmitting forces due to their flexible but tough mechanical properties. These properties are essential for the development of soft robotic joints that emulate human mechanics. Here, we have engineered humanoid tissues using PS toughened PIMs (Supplementary Note 3) that feature different polymer-inspired chain strain factors (f_c) enabling programmable mechanics under external stimuli and diverse engineering load conditions:

$$f_c = k_{f_c} \rho^{b_E} p^{d_E} (n_c^{c_E} - 1) \tag{1}$$

where b_E, d_E, c_E are experimentally calibrated exponents that quantify the sensitivity of the chain strain factor to molecular-like density, chain alignment, and crosslinking, respectively. k_{f_c} is the physical constant related to the properties of parent materials. The DIC results (Fig. 4a, Supplementary Fig. 8a, b, and Supplementary Movie 3) demonstrate that these humanoid tissues display a programmable strain response from stress stimulus, which escalated at higher levels of pre-stretching alignment. To tailor the mechanics of these humanoid tissues, we derived a framework for modulating the stress-responsive tissue modulus and f_c under different PS levels (Supplementary Fig. 8c). The elastic modulus of the tissue can be precisely programmed following a relationship outlined in Eq. 2, which is a derivation from the polymer-inspired chain strain from Eq. 1:

$$E = k_h f_c \tag{2}$$

where k_h is a fitting constant related to the property of parent material. Enabled by workings of tendons and muscles, we illustrate two typical mechanics in human joints: (1) a typical walk cycle enabled by the toe (Fig. 4b), characterized by three stages with distinct reaction slopes (k_1, k_2, k_3), and (2) the load responses of actuated (Fig. 4c) and passive (Supplementary Fig. 9) finger gestures with highly customized bending mechanics. Here, we programmed the PIM tissues to develop robotic components that can replicate human mechanics and are mechanically programmable. We developed PIM robotic foot joints with different gradients of polymer-inspired chain strain factors (∇f_c), where the higher ∇f_c resulted in less pre-stretching levels towards the edge of the joint and is expected to produce less stress reaction to the walk stimulus. This feature enables the programming of the humanoid toe mechanics. The robotic finger features a distributed polymer-inspired chain strain factor (f_c), tailored to emulate the bending reaction of a human finger. Mechanical experiments simulating toe flexion in a complete walking cycle indicate a successful mimicry of the human foot mechanics (Supplementary Fig. 10a, b, and also see Supplementary Fig. 10c for a printed robotic foot). Observations suggest the smaller reaction slope magnitudes in samples with higher ∇f_c aligning with our design expectations, and informing design guidelines for programmable load reaction in response to walking stimuli:

$$k_i = K_i \nabla f_c + A_i \tag{3}$$

where $i = 1$, 2, and 3 represents the indices of different reaction slopes, K_i and A_i are the experimentally derived parameters. Robotic humanoid fingers can be programmed as both actively actuated functional grabbing unit (Fig. 4c) or passive tissue joints being bent under external stresses. For active finger control, we developed a cable-driven robotic finger assembly, readily programmed through tailored distributions of different PIM joints. Experimental results confirmed that, by programming the f_c distribution within the cable-driven finger, it could be actively actuated into various gestures, enabling highly customizable grabbing tasks. For passive finger designs, experiments simulating various human finger bending gestures

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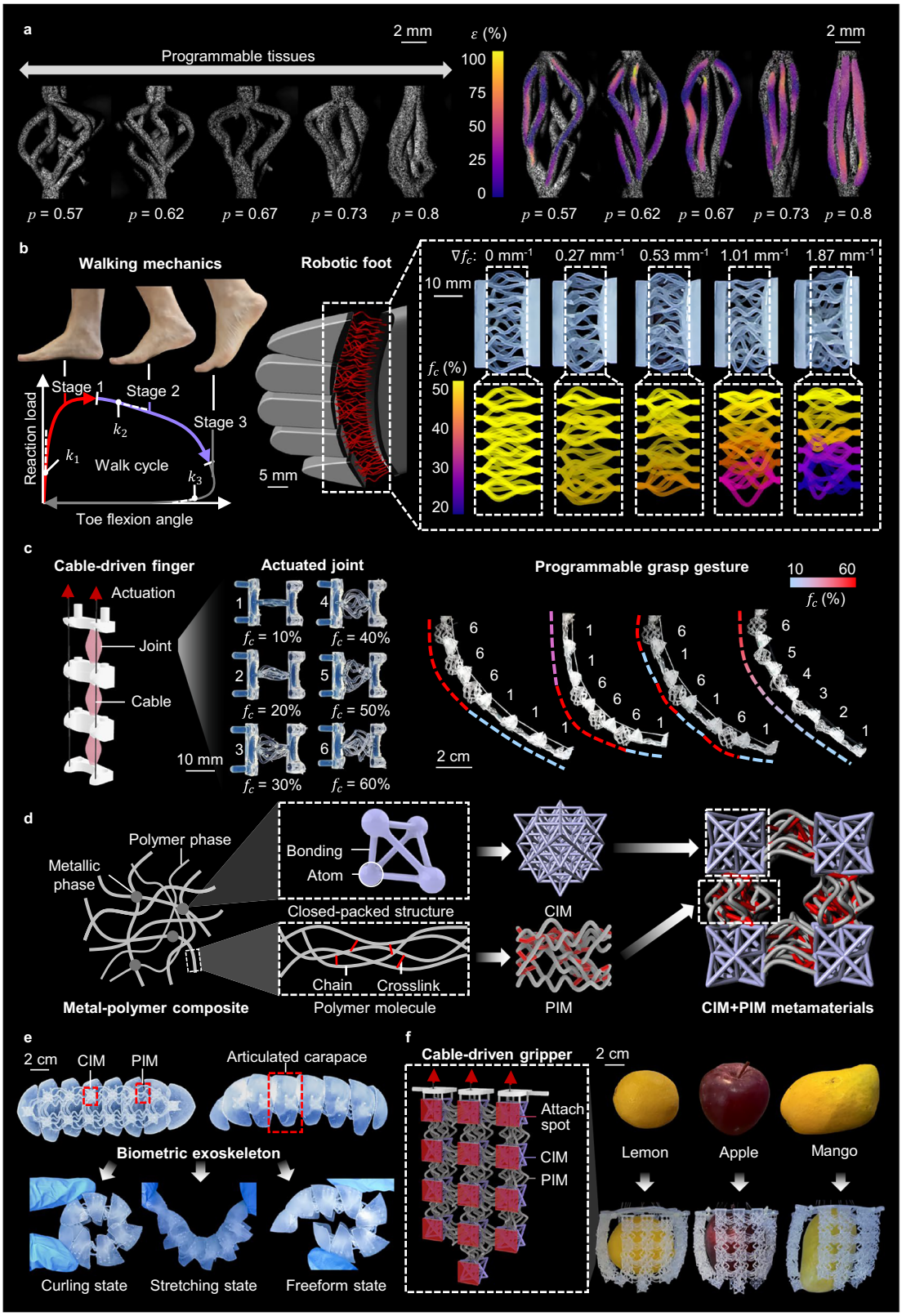


Fig. 4 | Programmable PIM humanoid tissue and functional applications. a The digital image correlation (DIC) analysis of the experimentally measured strain fields (right) for fabricated PIM humanoid tissues (left) with different pre-programmed PS levels. ε represents local DIC tissue strain. **b** Schematics of human walking mechanics and humanoid feet designed with different gradients of chain strain factors (∇f_c). The foot photos were photographed by author Zhenyang Gao. **c** Cable-driven finger schematics (left), fabricated joints (middle), and experimental captures (right) enabled by PIM humanoid tissues at actuated joints. Here, 1–6 represent different actuated joints with f_c engineered ranging from 10% to 60%, respectively. **d** Polymer + crystal lattice structure integration schematics. **e** An experimentally fabricated biomimetic exoskeleton with CIM+PIM offering highly deformable shapes while preserving its protective functions. **f** A cable-driven grab finger design schematics (left) and experiments (right) with CIM+PIM that can grab customized geometries.

(Supplementary Fig. 9) demonstrate that the robotic finger successfully replicated the bending reactions of a human finger. This response can be precisely programmed by adjusting the polymer-inspired chain strain factor (f_c) for each flexion (Supplementary Fig. 11). These results underscore the effectiveness of PIM and its programmable humanoid tissues, highlighting their potential for broad applications in biomimetic components that demand highly programmable and soft mechanics.

PIMs also demonstrated their capability to organically integrate diverse functional robotic and engineering structures in a programmable manner. For instance, composite materials incorporating metal particles within their polymer networks enhance mechanical properties while maintaining lightness, and can achieve anisotropically tailored behaviors by strategically distributing these metal particles. Hence, we implanted crystal-inspired octet-truss lattices into the polymer networks (Supplementary Note 4), and adjusted crosslink densities to create highly anisotropic CIM+PIM functional material sheets (Fig. 4d and Supplementary Fig. 12), while it should be noted that our method is not confined to octet-truss structures or sheet formats. These composite CIM+PIMs enable applications in soft biomimetic exoskeletons (Fig. 4e) and cable-driven robotic grab fingers (Fig. 4f). Cable-driven active deformation of the grabber is illustrated in Supplementary Fig. 13a, and the corresponding schematic is shown in Supplementary Fig. 13b. The grabber is twisted by cable to conform to the object geometry, thereby generating multiple holding points for secure grabbing, while grab tension and friction act together to prevent object slippage or escape. The CIM+PIM, with its variable crosslink densities, enhances a programmable specific modulus along the sheet compression direction ($\vec{p}$), while achieving an 85% reducible modulus at flexible stretch direction ($\vec{s}$) compared to conventional metamaterials (Supplementary Fig. 14a). It also exhibits a notable improvement in energy absorption capacity along $\vec{p}$ over its CIM counterparts (Supplementary Fig. 14b). Mechanical testing confirmed that the printed exoskeleton, featuring intricate carapaces interconnected with PIM and CIM under each carapace, responds with marked anisotropy to reshaping and shell-compression loads. This feature provides functional protection for soft robots, making them flexible and conformal for various functional modes as depicted in Fig. 1b. Additionally, the multimodal robotic finger, tested for soft bending and significant grab loads (Supplementary Fig. 15), demonstrates the capability to deform effectively into complex shapes while exerting sufficient grabbing force to handle objects, as illustrated in Fig. 4f.

In summary, we introduce mechanical metamaterials inspired by polymer molecular structures. The PIM approach is enabled by programming polymer-like twisted chains crosslinked by straight beams to translate polymer-inspired strengthening mechanisms (MD strengthening, PS strengthening, CL strengthening) to meta-structures in PIMs. The developed PIMs demonstrate combinations of high specific strength, specific modulus, energy absorption efficiency, and ductile fracture behaviors, expanding the range of properties that are currently affordable in metamaterials using crystal-like stiff architectures. Our findings suggest that the mechanical response of PIMs can be rationalized through an architecture-level analogue of polymer strengthening mechanics. By synergistically combining a crystal-inspired approach with the PIM one, we enable highly anisotropic crystal-polymer-composite metamaterials that could be used in soft robotics, such as a biomimetic exoskeleton and a versatile grab finger. By programming the microscale mechanics of PIM chains, advances have been made towards creating programmable humanoid tissues that successfully simulate human-tissue mechanical properties. This research not only introduces a class of mechanical metamaterials inspired by polymers but also provides a platform for exploring polymer-like strengthening through the pre-programmed architectural design.

Methods

Materials and sample preparation

In this paper, the Formlabs® Form 3 stereolithography printer is applied to fabricate the samples. The parent materials used in this study include Formlabs® clear v4, durable, and flexible resins, where the durable resins are selected as the base materials for the rest of the figures due to its pliable but stiff mechanics simulating the properties of the molecular chains. It should be noted that the proposed method is not limited to these types of materials. The majority of PIMs are 3D printed by stereolithography (SLA) using Formlabs® durable resin to mimic the high stretchability of each polymer chain, while the clear v4 and flexible 80 A resins are also used to systematically compare the mechanical improvement of PIMs for different parent materials. Experimental evidence confirms the incorporation of polymeric mechanics in metamaterials, characterized by molecular alignment, polymer-inspired chain scission, and extended fracture processes during the structural deformation. All samples are ultrasonically washed within absolute ethanol solutions for 10 min to remove adhered resin. The green clear v4 samples are tested without post-curing to avoid excessive brittleness. The durable samples are cured in a 60 °C ultraviolet environment for 60 min, while the flexible samples are cured in a 60 °C ultraviolet environment for 10 min instead. All samples are secured in a dark environment and tested within 1 h after printing or post-processing. The geometrical information of fabricated samples is summarized in Supplementary Table 1. The mechanical properties of the parent materials are evaluated based on the ASTM D638 standard, following the test configurations specified in the Methods following testing configuration summarized in Supplementary Fig. 17. Supplementary Table 2 further details the geometrical specifications and depicts various metamaterials, respectively. The diameters of the chains and bonding struts in PIMs, PIM-CIM composites, and PIM-based humanoid tissues have been standardized at 1 mm. Conversely, the diameters of the CIMs vary between 0.6 and 1.2 mm, facilitating a range of densities that overlap with those of the PIMs, enabling equitable comparisons in Ashby plots. The lower diameter limit was selected to ensure the printing fidelity of these intricate geometries. The geometry was discretized into voxels and evaluated layer by layer to identify unsupported or excessively overhanging regions. Designs containing potentially unprintable features were revised by increasing the local feature size or adjusting the structural orientation before fabrication.

Structural generation

The PIM architectures were generated using the polymeric structure generation workflow described in Supplementary Note 1 and Supplementary Figs. 22–24. The humanoid tissues, robotic toe and finger joints, biomimetic exoskeleton, and cable-driven grabber were subsequently designed and assembled in Rhino 6 without topology optimization or automated structural optimization. For the humanoid robotic components, PIM regions with programmed chain alignments and pre-stretch distributions were incorporated between rigid component regions to control the bending and actuation responses, as detailed in Supplementary Note 3 and Supplementary Fig. 16. For the CIM+PIM exoskeleton and grabber, octet-truss CIM regions were positioned at the load-bearing and attachment locations and interconnected by compliant PIM networks, following the designs presented in Supplementary Note 4 and Supplementary Fig. 12. The cable-actuation configuration of the grabber is further illustrated in Supplementary Fig. 13. The completed geometries were converted into printable solid STL models, exported for standard Formlabs® Preform slicing, and fabricated by stereolithography using a Formlabs Form 3 printer. No custom G-code or script-generated printing path was used. These functional components were printed using Formlabs durable resin and subjected to the same fabrication and post-processing conditions as the

PIM specimens, including ultrasonic washing in absolute ethanol for 10 min and UV post-curing at 60 °C for 60 min.

Mechanical experiments

Supplementary Figs. 17–20 summarizes the test configurations used in this study, employing a ZwickRoell Z100 universal tensile test machine. Supplementary Table 2 details the load rates, directions, and geometries for various mechanical tests satisfying quasi-static load conditions. ASTM D638 standard guidelines informed the mechanical evaluation of parent materials, with specimens conforming to ASTM D638 type IV (Supplementary Fig. 17). PIM samples featuring preprogrammed strengthening mechanisms and CIM designs were tested in a $20 \times 10 \times 20\ mm^3$ region along the x, y, and z axes, subject to vertical tensile or compressive forces (Supplementary Fig. 18). Tensile testing for CIM+PIM composite layers focused on polymer chain loading, employing specimens with two CIM+PIM cells aligned with the load direction and one cell perpendicular, reducing necking phenomena and enhancing the generalizability of mechanical properties (Supplementary Fig. 19a). Compressive tests for crystal loading mechanics used a single CIM+PIM cell perpendicular to the load direction (Supplementary Fig. 19b). Biomimetic shells underwent tensile and compressive testing to assess reshaping and compression mechanics (Supplementary Fig. 19c, d). A single grabbing finger was tested for both stretching (grab force) and bending (adapting force) (Supplementary Fig. 19e, f). Cylindrical tissue test specimens with a molecular-like density of $4.25\ cm^{-3}$ under different PS levels were designed to evaluate tissue mechanics (Supplementary Fig. 20a). For robotic toe and finger joints, specialized fixtures simulated bending loads akin to human tissue dynamics (Supplementary Fig. 20b–d).

The experimental stress–strain curves are used to calculate the modulus, strength, energy absorption efficiency, fracture dissipation, and fracture energies of the specimens. The energy absorption efficiency is calculated by integrating the compressive stress–strain curve till the densification strain, normalized by the peak stress (Eqs. 4 and 5). The fracture energies are evaluated by integrating the tensile stress–strain curves till the total fracture of the specimen, while the fracture dissipation integrates the curves from the fracture initiation strain to the complete fracture (Eqs. 6 and 7). The specific mechanical properties are derived by normalizing these mechanical performances by the density of the specimen within the test region to conduct fair comparisons.

$$\eta = \int_0^{\varepsilon_d} \sigma \varepsilon d\varepsilon / \sigma_{peak} \tag{4}$$

$$\sigma_{peak} = MAX(\sigma(\varepsilon)),\ \varepsilon \in [0,\ \varepsilon_P], \tag{5}$$

$$A_{fracture} = \int_0^{\delta_b} F\, d\delta, \tag{6}$$

$$A_{dissipation} = \int_{\delta_a}^{\delta_b} F\, d\delta \tag{7}$$

where η is the energy absorption efficiency, σ is the compressive stress, σ_{peak} is the peak stress, ε is the compressive strain, ε_P is the plateau initiation strain, $A_{fracture}$ is the fracture energy, $A_{dissipation}$ is the fracture dissipation energy, F is the fracture load, δ_a and δ_b are the displacement at the fracture initiation and complete fracture, respectively.

Digital image correlation analysis

The DIC analysis is performed on a Correlated Solutions VIC-3D DIC measurement system using its 2D measuring configurations (Supplementary Fig. 21). The DIC is used to measure full-field surface strains during mechanical testing of PIM samples. These spatial strain maps were used to locate regions associated with different strengthening events (entangled zones, crosslinked segments, and proto-crystalline chains) and quantify the relationship between experimental local chain strain and the programmed PIM design, such as robotic muscles. The speckle patterns are sprayed evenly on the specimens for DIC calculations. A Basler acA4112-30 um camera with 28-mm Basler lens are used to capture the relative positions of the DIC speckle patterns, while VIC-2D 7 software is used to perform DIC analysis based on the experimental results. The analyzed images possess a resolution of 4096 pixels × 3000 pixels. A 19 pixels × 19 pixels subset dimension is used during the DIC derivation, while the step size is 4 pixels.

Data availability

The data supporting the findings of this study are included within the paper and its Supplementary Information. All other data are available from the corresponding authors upon request. Source data are provided with this paper.

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Author contributions

Z.G. conceptualized this study. Z.G., H.Z.W., M.S.P., and Y.L. contributed to the methodology design of the study. Z.G., Y.D., and P.R. conducted simulations. Z.G. and S.W. performed the experiments. Z.G. performed data analysis, interpretation, curation, and wrote the manuscript. Z.G., H.Z.W., M.S.P., Y.L., Y.W., P.R., G.Z., X.S., S.W., K.Z., Y.Z., Y.B.L., L.K., H.J., L.Z., S.Y., T.F., Z.T., and H.S. substantially revised the work. Y.W., H.Z.W., Y.Z., and H.W.W. administered the project and provided supervision throughout the study.

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Competing interests
The authors declare no competing interests.

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